



Full length article

Escape behaviors of the Eurasian Tree Sparrow (*Passer montanus*) across China: Northern populations are bolder than southern populations

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ABSTRACT

Urbanization has significantly altered the habitat structure and behavioral patterns of animals. In urban environments with frequent human disturbances, animals may undergo adaptive adjustments in their escape behaviors. This study focuses on Eurasian Tree Sparrows (*Passer montanus*) from five cities across different latitudes in China, comparing their flight initiation distance (FID) in urban and rural areas to assess the impact of urbanization on their escape behavior and the trend of ecological homogenization. The results confirm the established pattern at the geographic level: a reduction of FID in urban habitats and a decrease in FID with increasing latitude. That is, northern populations of the Eurasian Tree Sparrow in China are bolder than the southern populations. In addition, FID is positively correlated with flock size, with rural flocks exhibiting more pronounced group behavior (observed flocking frequency for urban and rural is 28.74% vs. 40.7%, flock size variance is 26.90 vs. 55.63). Despite the differences between northern and southern latitudes, the variability of FID in urban individuals consistently remains lower than that in rural areas, supporting the trend of reduced variability in escape behavior among birds in urban environments. Driven by urbanization, the escape behavior of Eurasian Tree Sparrows tends toward ecological homogenization, meaning that behavioral differences between urban areas in different cities are diminishing. This trend may obscure the behavioral variability brought about by latitude gradients, indicating that urbanization not only shapes the behavioral adaptations of birds but also potentially weakens their behavioral diversity.

1. Introduction

In a natural world, behavioral adaptations are crucial for the survival and reproduction of animals. However, with the rapid acceleration of urbanization, the scope and intensity of human disturbances have increased, causing notable impacts on wildlife. As a result, the anti-predator behavior exhibited by animals in response to human presence has also changed. Understanding and mitigating the effects of human activities on wildlife behavior are essential for promoting harmonious coexistence between humans and animals (Sol et al., 2013). In an early experiment, Walther (1969) found that Thomson's Gazelles (*Eudorcas thomsonii*) responded similarly to vehicles and predators. Based on this study, later research suggested that non-lethal human disturbances function as perceived predation risks (Frid and Dill, 2002), indicating

that animals react to humans in a manner similar to their response to predators.

The classic optimal escape theory states that animals weigh the costs and benefits of escape in the face of a threat to decide when to escape (Ydenberg and Dill, 1986). This theory suggested that during vigilance, animals rapidly assess the risks of remaining in place against the energetic costs of fleeing, thereby forming an optimal escape strategy suited to their physical condition. Premature escape may reduce foraging efficiency or increase physiological costs, whereas delayed escape may elevate the risk of predation-induced mortality (Ydenberg and Dill, 1986). This theory provides an important theoretical basis for understanding changes in flight initiation distance (FID) in birds.

Escape behavior in birds is often measured by FID, which refers to the distance at which an individual begins to flee when approached by a

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potential predator, usually a human (Tarlow and Blumstein, 2007). This metric reflects an economic decision-making process in birds. Therefore, FID can indicate the capacity of birds to coexist with humans (Møller, 2008). For urban-dwelling birds, human approach represents a non-lethal and low-risk event, as humans in cities rarely pose a direct threat through hunting or other means (Berger, 2007; Yuan et al., 2024; Liu et al., 2025a). Consequently, birds that gradually adapt to human disturbances often exhibit reduced wariness towards human proximity. Numerous studies have shown that birds in areas of frequent human contact tend to better tolerate human presence and exhibit reduced FID (Cooke, 1980; Webb and Blumstein, 2005; Linn et al., 2015; Samia et al., 2015). In addition, changes in bird tolerance to humans are not only behavioral adjustments within individuals, but may also involve evolutionary adaptations at the population level. At the same time, the differences in FID of birds in different regions also reflect species-level selection, with adaptive species more likely to occupy human-dominated habitats (Mikula et al., 2023).

Cities are designed to meet human needs, leading to a high degree of physical similarity across urban environments worldwide (McKinney, 2006). This similarity allows cities to be considered as “environmental filters”, where only species with certain traits can survive. Therefore, ecological homogenization due to urbanization has been a widely discussed topic (Gronoffman et al., 2014). However, previous studies have mainly focused on urban biotic homogenization (McKinney, 2006). Some researches have examined biodiversity, noting that urbanization leads to increased similarity in species composition across cities (Devictor et al., 2007; Suarez-Rubio et al., 2011). Others have investigated homogenization in species’ external traits. For example, Leveau (2019) found that birds in highly urbanized areas predominantly exhibit gray plumage, a phenomenon referred to as plumage color homogenization. Additionally, studies have suggested that birds capable of successfully adapting to urban environments tend to have large brains and exhibit great foraging innovation (Sol et al., 2005; Maklakov et al., 2011). Moreover, food provisioning—a ubiquitous feature of urban landscapes, can also significantly reduced the FID of urban birds (Møller et al., 2015), providing complementary explanation for the urban-rural FID divergence. While these studies have explored the genetic, functional, and evolutionary impacts of urban homogenization on biological communities, little research has investigated whether birds’ escape behavior also undergoes ecological homogenization due to urbanization.

In studies on avian escape behavior, considerable research has examined latitudinal patterns in predation and escape responses (Vermeij, 1978; Diaz et al., 2013; Andrade and Blumstein, 2020). Additionally, numerous studies have compared FID between urban and rural environments, consistently showing that birds in urban centers exhibit shorter FID than those in rural areas (Samia et al., 2015, 2017; Davey et al., 2019). However, little research has explored whether birds’ escape behavior exhibits urban ecological homogenization and how FID varies between urban and rural areas across different latitudes. Therefore, in this study, we collected FID data of Eurasian Tree Sparrows (*Passer montanus*) from urban and rural areas of five cities across China. Our goal was to examine how FID varies across different latitudes and whether urbanization drives ecological homogenization in birds’ escape behavior. Based on previous studies, we predicted that the FID of Eurasian Tree Sparrows in Chinese cities would follow a latitudinal gradient, with FID in urban areas shorter than in rural areas. Urban birds have a wide ecological niche (Evans et al., 2011; Marzluff, 2012) and a broad environmental tolerance (Bonier et al., 2007). As a common urban bird, Eurasian Tree Sparrows are also widely distributed in China and can adapt to different environments. Therefore, we hypothesized that Eurasian Tree Sparrows would not exhibit an ecological homogenization effect in their FID.

2. Materials and methods

2.1. Study area and study species

The capital city of each of the five provinces (or municipalities) in China was chosen as a study site, as capital cities tend to be more developed, have higher levels of urbanization, and host larger permanent populations. The five cities, listed in order of latitude variation from high to low latitude, were Changchun City, Jilin Province (43°05′–45°15′ N, 124°18′–127°05′ E), Beijing (39°26′–41°03′ N, 115°25′–117°30′ E), Zhengzhou City, Henan Province (34°16′–34°58′ N, 112°42′–114°14′ E), Hangzhou City, Zhejiang Province (29°11′–30°33′ N, 118°21′–120°30′ E), and Haikou City, Hainan Province (19°31′–20°04′ N, 110°07′–110°42′ E) (Fig. 1). According to the geographical division of China, Changchun, Beijing, and Zhengzhou are classified as northern cities, while Hangzhou and Haikou (including Lingshui) are classified as southern cities.

We defined the urban areas of each city as the central districts, whereas rural areas were located at the city’s periphery, where farmland areas were present. Each city had one designated urban and one rural sampling site. During our field surveys, we found that Eurasian Tree Sparrows had a limited distribution and low population density in Haikou City, making it difficult to collect sufficient rural samples there. To address this, we classified all data collected in Haikou as urban data and selected Lingshui County (18°21′–18°47′ N, 109°45′–110°08′ E), which is located some distance from Haikou, as the corresponding rural area for comparison.

The Eurasian Tree Sparrow, as one of the earliest species associated with human urban life (Møller, 2015), forages mainly on the ground and often utilizes urban resources. As a dominant urban species, it serves as an ideal model for studying the effects of urbanization on birds. Furthermore, its widespread distribution and resident status in China (Zheng, 2023) facilitated data collection, making it a suitable focal species for this study.

2.2. Data collection

As the FID of birds can be influenced by the attire of observers (Feng and Liang, 2020a; Zhou and Liang, 2020), all observers in this study wore identical clothing: black tops and trousers, dark-colored shoes, and had black hair. Field surveys were conducted during the non-breeding season in October 2024. Data collection took place during peak bird activity hours, from 07:30 to 11:30 a.m. in the morning and 02:30 to 04:30 p.m. in the afternoon. We chose to approach individuals with natural state, such as perching, foraging, or preening, whereas highly vigilant or obviously alert individuals were excluded. In the case of flocks, we measured the FID of the first individual to flee. Observers used a rangefinder binocular [SNDWAY SW-M500, accuracy: 0.1 m; Sndway Technology (Guangdong) Co., Ltd, Dongguan, China] and approached the focal bird at a constant speed of 0.5 m/s in a direct path.

We employed a standardized approach where observers avoided direct eye contact with the target individual and slowly approached to a fixed distance of 30 m. At the 30-m mark, observers briefly paused to confirm that the bird was in a natural state before standing still momentarily to begin the FID measurement. The observer stopped moving as soon as the bird took flight, and the FID was recorded (Blumstein, 2006). To minimize external disturbances, FID measurements were conducted in locations with no other people or moving objects nearby, ensuring that the bird’s flight response was solely because of the observer’s approach. In most cases, this was chosen as a fixed distance of 30 m to avoid a correlation between FID and starting distance. If birds started moving when the observer was longer than 30 m away, a 50 m was used (this amounted to <0.002 % of all observations).

In addition, we found that there was some clustering of Eurasian Tree Sparrows, so the number of individuals within a 10 m radius of the focal

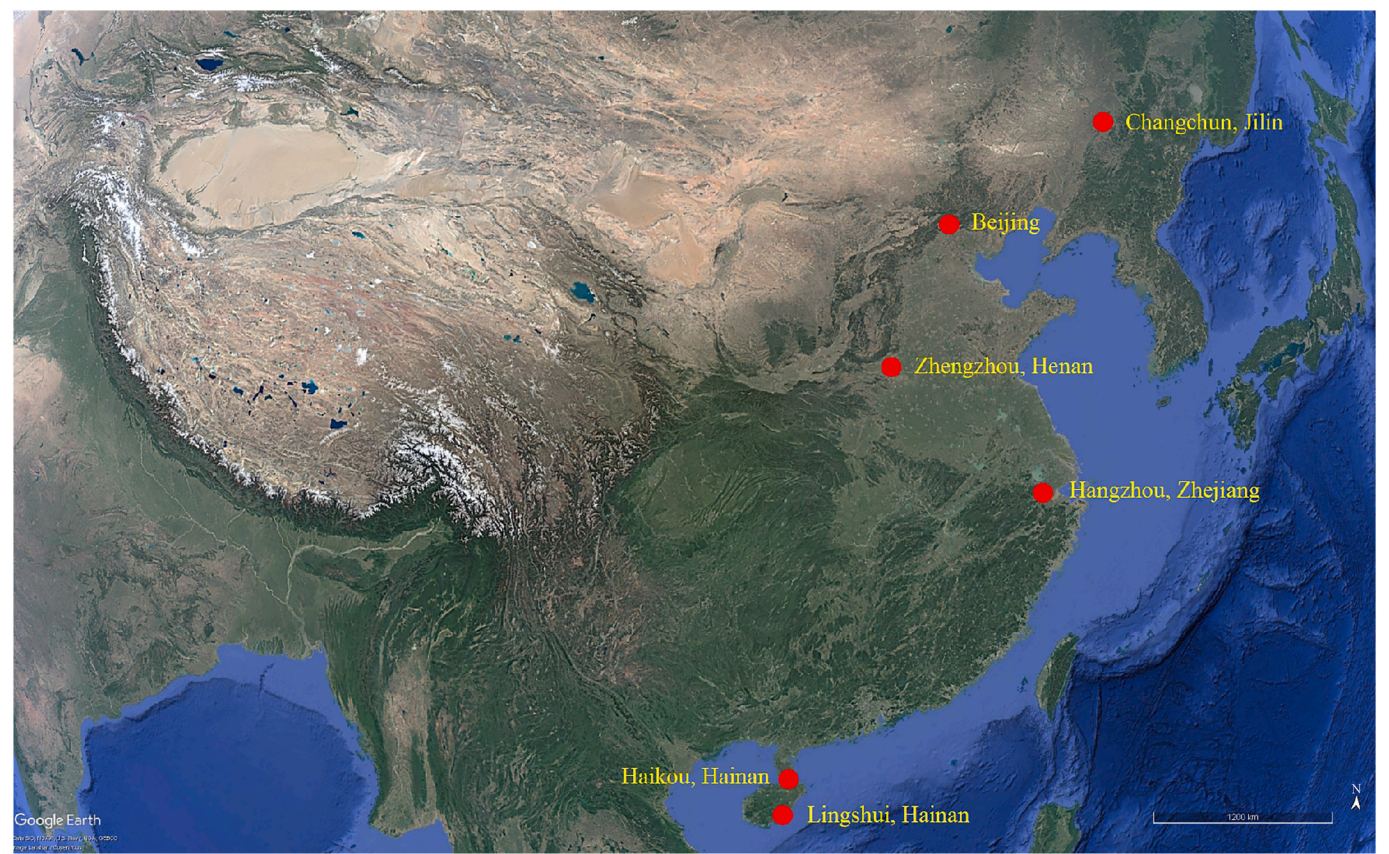


Fig. 1. Sites of six study sites and five provinces for which FID data were collected in China. The map does not show the specific locations of urban and rural areas, but the urban and rural locations we selected in each city are adjacent. Except for Hainan Province, Haikou was designed as the urban area, while Lingshui served as the rural area.

animal was recorded as the flock size, and we considered the number of individuals three or more as a flock. The experimental methods we used did not harm the birds and we tried to avoid interference from other people during the collection process. We moved at least 100 m away after recording one data point before selecting another focal bird.

2.3. Data analysis

Data were analyzed in R software (R Core Team, 2023). A Shapiro-Wilk test was conducted to assess the normality of the data, revealing that the data did not follow a normal distribution. The Levene test was employed to examine variance differences between different cities and places (and their interactions). Consequently, a Kruskal-Wallis rank sum test was used to evaluate the differences in FID between urban and rural areas across the five cities. Additionally, to further verify the presence of homogenization among cities, we grouped the three northern cities and the two southern cities separately. Then we calculated the Coefficient of Variation (CV) for the FID data in both two areas to evaluate the variability of FID within different regions.

In addition, A generalized linear mixed model (GLMM) with a Gamma distribution and log link function was applied to further analyze the data. In this model, city, flock size, and place (urban or rural area) were included as fixed effects, whereas the experimental person was incorporated as a random effect to account for individual differences among data collectors. The model was fitted using the glmer function from the {lmerTest} package (Kuznetsova et al., 2017), and used “bobyqa” to optimize the model. Data were presented in the form of mean ± standard deviation (Mean ± SD).

3. Results

We collected FID data from Eurasian Tree Sparrows in urban and rural areas of five cities, resulting in a total of 1819 valid FIDs. In Changchun, 160 data were collected in urban areas and 226 in rural areas. In Beijing, 152 data were collected in urban areas and 247 in rural areas. In Zhengzhou, 209 data were collected in urban areas and 153 in rural areas. In Hangzhou, 108 data were collected in urban areas and 299 in rural areas. In Haikou, 46 data were collected in urban areas and 219 in rural areas. The variability of FID in rural areas is much greater than in urban areas (Levene’s test: $F = 34.8$, $df = 1$, $P < 0.001$). In both urban and rural areas, the average FID of Eurasian Tree Sparrows increases as latitude decreases, and the differences in FID between urban and rural areas across the five cities are highly significant ($P < 0.001$, Kruskal-Wallis rank sum test; Table 1). Additionally, we found that in northern cities, the CV is 0.498 for urban areas and 0.552 for rural areas;

Table 1
Comparative analysis of FID data for urban and rural areas across five cities using the Kruskal-Wallis rank sum test.

City	Site	N	FID (m)	P
Changchun	Urban	160	3.99 ± 2.02	<0.001
	Rural	226	8.41 ± 3.79	
Beijing	Urban	152	4.16 ± 2.07	<0.001
	Rural	247	11.9 ± 6.27	
Zhengzhou	Urban	209	4.31 ± 2.12	<0.001
	Rural	153	12.4 ± 6.83	
Hangzhou	Urban	108	6.41 ± 2.78	<0.001
	Rural	299	13.6 ± 6.18	
Haikou	Urban	46	6.48 ± 2.02	<0.001
	Rural	219	14.3 ± 5.23	

in southern cities, the CV is 0.400 for urban areas and 0.417 for rural areas (Fig. 2).

The data collectors had minimal impact on FID measurements ($P_{\text{variance}} < 0.001$, GLMM; Table 2). As Changchun has the highest latitude, we used Changchun as the intercept in our model. We found differences in FID among cities, with FID increasing relative to Changchun. Moreover, the FID had a tendency to increase as latitude decreases (Fig. 2). In addition, in China, demarcated by the Qinling-Huai River Line dividing north and south, we find that there is no significant difference between the three urban areas in the north (Changchun, Beijing and Zhengzhou, $P > 0.999$) and between the two urban areas in the south (Hangzhou and Haikou, $P > 0.999$), but marked differences emerged between urban areas across northern and southern regions ($P < 0.001$). This pattern was absent in rural habitats. Regarding flocking behavior, urban Eurasian Tree Sparrows were less likely to form flocks than their rural counterparts, but also significantly reduced variation in flock sizes when aggregated. In our study, observed flocking frequency was defined as the percentage of total observations recording ≥ 3 birds in association. This standard metric is calculated as: observed flocking frequency = (number of flock-containing observations/total observations) $\times 100\%$. In urban areas, the observed flocking frequency was 28.7% ($N = 194$ out of 675 total observations), whereas in rural areas, the observed flocking frequency was 40.7% ($N = 466$ out of 1144 total

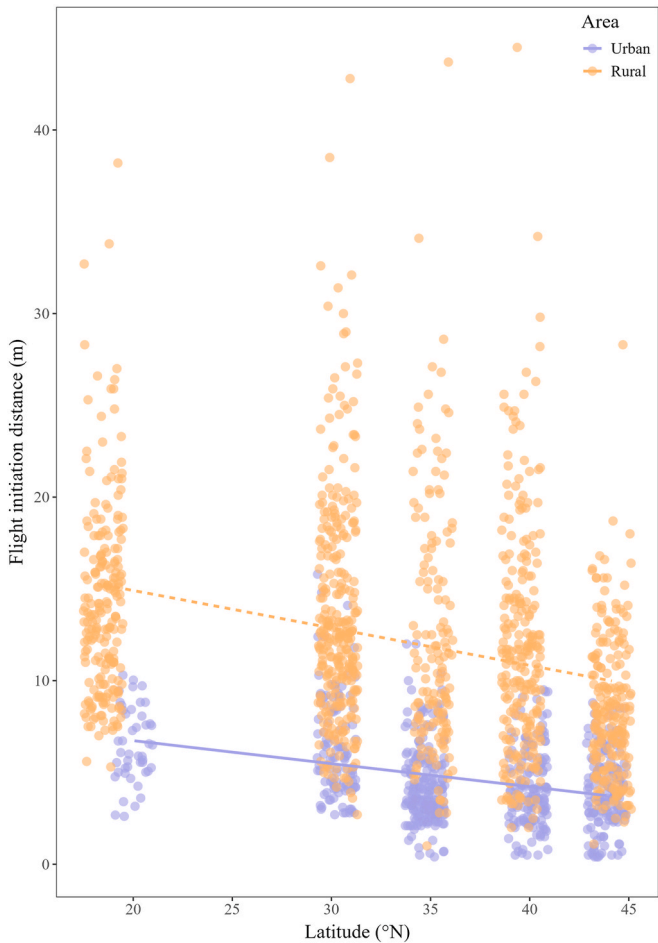


Fig. 2. Variation in FID across latitudes. Data points represent paired urban and rural sites, with each data point shown as a circle. The straight line represents the linear regression based on the average values. To enhance clarity, the actual latitude points for urban and rural sites are slightly offset. The average FID values in both urban and rural areas of the five cities show a declining trend, with the decline being more pronounced in rural areas. Additionally, FID variability is greater in rural areas than in urban areas.

Table 2
Generalized linear mixed model (GLMM) was applied to analyze the FID data collected from five cities, evaluating the effects of city, sampling site, population size, and sampling personnel on the FID of the Eurasian Tree Sparrow.

Group	Name	Variance	Standard deviation (SD)
Experimental person	(Intercept)	0.000,388,8	0.019,72
Residual		0.240,771,2	0.490,68

	Estimate	Standard Error	t-value	P
(Intercept)	0.388,209	0.050,165	7.739	<0.001
Beijing	0.203,074	0.046,976	4.323	<0.001
Haikou	0.569,797	0.084,080	6.777	<0.001
Hangzhou	0.442,511	0.038,208	11.582	<0.001
Zhengzhou	0.214,585	0.046,119	4.653	<0.001
Site	0.893,359	0.026,968	33.127	<0.001
Flock size	0.007,618	0.001,910	3.988	<0.001

observations). The variance of urban flock size is 26.90, the variance of rural flock size is 55.63, and the urban/rural variance ratio is 0.4836. Additionally, flock size had a significant effect on FID, with FID increasing as flock size increased (estimate = 0.007, $t = 0.988$, $P < 0.001$, GLMM; Table 2).

4. Discussion

In this study, we measured the flight initiation distance (FID) of Eurasian Tree Sparrows in urban and rural areas of five cities across China. We found that the variability of FID in rural areas was significantly greater than in urban areas. Although there were significant differences in FID among the cities, these differences were mainly associated with latitude, with FID increasing as latitude decreased. This finding supports our initial hypothesis that birds in urban areas exhibited urban homogenization in their escape behavior towards humans.

All five cities, urban Eurasian Tree Sparrows had lower FID than rural ones. This may be because urban birds face reduced predation risks (Lepczyk et al., 2003; Møller and Ibáñez-Álamo, 2012; Samia et al., 2017). According to the economic escape theory, can optimize their energy allocation by delaying escape, and invest more resources in key activities such as foraging and reproduction (Ydenberg and Dill, 1986; Cooper and Frederick, 2007). Human activities have long been recognized to influence avian tolerance, often manifesting in a well-established urban–rural gradient. Our study provides geographic validation of this widely documented pattern within the Chinese context: urban birds exhibit significantly shorter FID, while northern populations tend to display higher levels of tolerance. These findings align closely with the theoretical framework of the “boldness gradient” in global urban ecology, offering new empirical evidence for behavioral adaptation across cultural and geographic regions (Samia et al., 2017). We believe that the underlying mechanism of this urban–rural divergence is primarily driven by the difference in human footprints: urban areas, despite high human presence, pose lower actual threats and offer stable food resources, thereby promoting reduced wariness through behavioral plasticity; in contrast, human disturbance rural areas are usually sporadic, maintaining more conservative anti-predator responses (Carrete and Tella, 2013; Feng and Liang, 2020b; Mikula et al., 2023; Liu et al., 2025b).

A unique pattern of FID distribution was noted between cities in northern and southern China: no significant differences were detected between urban areas in the three northern cities, and there was also no statistical difference between urban areas in the two southern cities. This pattern likely arises from differences in park habitat structures. The northern urban parks generally show a vegetation pattern dominated by low water coverage and hardy trees (such as poplar and pine), forming open space permeability. In contrast, south park is characterized by high water network density and abundant subtropical evergreen vegetation

(such as *Ficus* and Lauraceae), and its regional plant community structure leads to reduced visual permeability (Liu, 2011; Zhang and Zhang, 2024). We also observed that FID decreases with increasing latitude, a trend consistent with previous studies (Díaz et al., 2013; Blumstein et al., 2016; Samia et al., 2017), and this trend is observed in both urban and rural areas of the five cities. Several factors might explain this phenomenon. First, biodiversity typically decreases with increasing latitude, and birds at higher latitudes face fewer predators and more limited food resources (Rohde, 1998). Consequently, birds may increase their tolerance of potential predators, including humans, to access scarce food resources more efficiently (Liu et al., 2025b). Second, our study was conducted in the Northern Hemisphere during the fall, a time when temperatures in three cities at high-latitude regions have already begun to drop, whereas those in two cities at low-latitude regions remain relatively high. Under these climatic conditions, birds at high latitudes require more food to maintain their body temperature, which may motivate them to exhibit higher levels of bolder behavior and take greater risks in searching for food resources (Crouchet et al., 2024).

However, explaining the regional consistency in the distribution of FID between northern and southern cities in China solely from an ecological structure perspective remains insufficient. This behavioral convergence phenomenon may reflect a deeper characteristic of “institutional homogenization” within Chinese urban ecosystems. Chinese cities exhibit a high degree of consistency in planning concepts, architectural styles, public green space design, and human daily behavioral patterns, which is unique globally due to standardized urban construction led by the state (Wu, 2013). In contrast, European cities have a longer history of urbanization, with more diverse urban forms, cultural backgrounds, and ecological interactions (Xie, 2016). European urban green spaces typically retain more natural elements, and interactions between residents and wildlife are more personalized and diverse (Díaz et al., 2013). This difference is not only reflected in urban design but may also profoundly influence how birds perceive risk from humans. Some European cities, such as those in the UK, have long promoted the ecological concept of “coexistence with nature”, where citizens show greater tolerance towards wildlife, even actively feeding birds (Jones, 2011). In contrast, in China, despite frequent human activity, interactions between humans and wildlife are more passive and non-interventionist. Furthermore, the urbanization process in Chinese cities has been concentrated in recent decades, with urban green spaces mostly being artificially constructed, lacking continuity of original habitats, and with a steeper ecological gradient in the urban-rural transition zones.

At the same time, China’s unique sociohistorical trajectory may have further reinforced this pattern. The nationwide “Four Pests Campaign” during the 1950s–1960s involved indiscriminate extermination of sparrows (You, 2000), potentially resulting in genetic bottlenecks that could affect behavioral plasticity. Subsequently, rapid urbanization and increasing landscape homogenization have accelerated behavioral adaptation among urban populations. The expansion of modern Chinese cities has created highly standardized residential environments, where birds frequently encounter non-threatening human activity in predictable settings such as parks and housing complexes. In contrast to the sporadic and defensive human-wildlife interactions more typical in rural areas, these highly spatiotemporally predictable urban environments are more conducive to the development of tolerant behaviors.

In Chinese cities, the escape behavior of birds in response to human disturbance may not only reflect an adaptation to the urban physical environment but also a behavioral response to a highly predictable and standardized human social behavior. We propose that the influence of urban homogenization could partially obscure behavioral variation driven by latitude gradients. Future research should integrate historical population data with modern genomic analyses to quantify the long-term effects of socio-historical events on behavioral plasticity and adaptive evolution, further elucidating the complex mechanisms of animal behavioral adaptation in the context of urbanization.

Additionally, our study found a positive correlation between FID data and flock size. This result supports the early-warning hypothesis (Lazarus, 1979; Morelli et al., 2019), which states that animals in large flocks detect predators earlier, and the larger the flock, the earlier they begin to flee. Alert individuals in a flock will be the first to flee, which triggers a chain reaction within the flock that drives other individuals to fly in alarm (Kahlert, 2006; Lehtonen and Jaatinen, 2016). In our experiments, we found that Eurasian Tree Sparrows living in rural areas were more willing to form flocks, and the variance of flock size in rural areas is larger. Conversely, the urban landscape imposes two critical social constraints: reduced flocking frequency and diminished flock size variation. This may be due to the increased number of predators in rural environments and the enhanced risk of predation (Lepczyk et al., 2003; Møller and Ibáñez-Álamo, 2012; Samia et al., 2017).

In conclusion, our findings reveal a notable effect of urbanization and ecological homogenization on the escape behavior of Eurasian Tree Sparrows. Urban individuals exhibited a significantly higher tolerance toward humans compared to their rural counterparts, and their FID showed a decreasing trend with increasing latitude. More importantly, we observed a clear pattern of ecological homogenization in escape behavior due to urbanization. This suggests that urbanization not only influences behavioral patterns in birds but also, to some extent, reduces behavioral differences across geographic regions, leading to more similar escape responses among birds in different cities. It is worth noting that although urban homogenization may help animals adapt to urban environments in the short term, this process is leading to a weakening or even disappearance of escape behaviors, potentially at the cost of the species’ original behavioral diversity (Sol et al., 2013).

CRediT authorship contribution statement

Yidong Wei: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Yuran Liu:** Writing – original draft, Methodology, Investigation, Formal analysis. **Kangning Luo:** Methodology, Investigation. **Qiqi Liu:** Methodology, Investigation. **Fangfang Zhang:** Methodology, Investigation. **Weihui Xing:** Methodology, Investigation. **Caizhen Wen:** Methodology, Investigation. **Jinmei Liu:** Writing – review & editing, Validation, Methodology, Investigation, Data curation, Conceptualization. **Wei Liang:** Writing – review & editing, Validation, Software, Funding acquisition, Conceptualization.

Ethic statement

The experiments comply with the current laws of China. No special permit was required for this study as it was not involved in animal or plant collection.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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